Name: Trinh Gia Hiep - 2352343 - HW3

1. Input with bias: $x = (x_0, x_1, x_2) = (1, 1, -1)$

$w = (0,1; -0,3; 0,4)$, $y = -1$

* $z = w.x = 0,1.1 + (-0,3).1 + 0,4.(-1)$

$= -0,6$

- Prediction $\hat{y} = \text{sign}(-0,6) = -1$

$\rightarrow \hat{y}$ = target so the example is classified correctly so no update needed.

2. sigmoid $\sigma(z) = \frac{1}{1+e^{-z}}$

z	ReLU = max(0,z)	Sigmoid	Calculation
-3	0	0,0474	$1/1+e^{3} \approx 0,047$
-0,5	0	0,3775	$1/1+e^{0,5} \approx 0,378$
0	0	0,5	$1/1+e^{0} = 0,5$
2	2	0,8808	$1/1+e^{-2} \approx 0,881$

- Differences

Aspect	ReLU	Sigmoid
Output range	$[0, \infty)$, unbounded above	$(0,1)$ bounded
Saturation	Does not satured for $z > 0$ (derivative = 1)	Satures at both ends
Gradient	Exactly 0 for $z < 0$, 1 for $z > 0$.	Never exactly 0 but vanishes.

- tanh vs sigmoid: tanh is preferred when we want a zero-centered hidden activations. Sigmoid output are always positive, so all weight gradients into the next layer share the same sign, may causes zig-zag updates.

3. $x = 1$, $w = 0,6$, $v = -0,7$, $t = 0$, bias $= 0$, $L = 0,5(t-0)^2$

- forward Pass:

$z_h = w.x = 0,6.1 = 0,6$

$h = \sigma(0,6) = \frac{1}{1+e^{-0,6}} = \frac{1}{1+0,5488} \approx 0,6457$

$z_0 = v.h = -0,4520$

$o = \sigma(-0,4520) = \frac{1}{1+e^{+0,4520}} \approx 0,3889$

$L = 0,5(0-0,3889)^2 = 0,0756$

- Outer-layer error signal:

$\delta_0 = \frac{\partial L}{\partial z_0} = (o-t).o(1-o)$

$o(1-o) = 0,3889.(1-0,3889) = 0,2377$

$\rightarrow \delta_0 = 0,3889.0,2377 = 0,0924$

- Gradient for v: $\frac{\delta_L}{\delta_v} = \delta_0.h = 0,0924.0,6457$

$= 0,0597$

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⇒ Positive gradient, so gradient descent decreases v, making it more negative.

3. Gradient for w

- $\delta_h = \delta_o . v . h(1-h) = 0,0924 . (-0,5) . 0,6457 . 0,3543$

$= -0,0148$

- $\frac{\delta L}{\partial w} = \delta_h . x = -0,0148$

⇒ Negative gradient, so G.D increases w.

4. Sketch description:

- Undersitting: Two nearly overlapping curves stay high and flat.
- Healthy: Two curves that fall together and settle close to each other at a low level.
- Oversitting: the curves diverge, the training curve keeps falling while the validion curve bottoms out and climbs.

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5 Prediction:

- Train score : A ≈ 0,98, B ≈ 0,99
- Validation score : A ≈ 0,9, B ≈ 0,91
- Runtime : B is slower.

Results:	A:	B
Train score	1,0	1,0
Validation score	0,9274	0,9324
Runtime	21,4s	29,4s

It does support my predictions (mostly).